

Day - 2: Architecture Deep Dive

▼ LPC2129 Microcontroller | LPC Microcontroller Architecture

1. Introduction & General Specifications

- **Microcontroller:** LPC2129.
- **Manufacturer:** NXP (founded by Philips).
- **Architecture Base:** It is a 32-bit microcontroller based on the ARM7 architecture.
- **Package Type:** It is an LQFP (Low Profile Quad Flat Package) IC. This means it has pins on all four sides and operates at a low profile voltage of 3.3 volts.
- **Pin Count:** Features 64 total pins, with each pin capable of multiple functions.

2. Processor & Clock Management

- **Processor Core:** Uses the ARM7TDMI-S processor.
 - *T = Thumb mode, D = Debug mode, M = Multiplier, I = In-circuit emulation, S = Synthesizable version.*
- **Operating Frequency:** The processor can operate at a maximum frequency of 60 MHz.
- **Clock Generation:**
 - It uses an external crystal oscillator of 12 MHz.
 - A **PLL (Phase Locked Loop)** circuit is used to step up or reduce the CPU frequency from the base 12 MHz crystal.

3. Bus Architecture (AMBA)

The microcontroller uses **AMBA** (Advanced Microcontroller Bus Architecture) to connect the CPU (system block) with the internal peripherals (peripheral

block).

- **Clock Division:** AMBA is responsible for dividing the main clock between the CPU and peripherals using an APB divider.
- **C Clock (CPU Clock):** Runs the system block (e.g., at 60 MHz).
- **P Clock (Peripheral Clock):** Runs the peripheral block, typically operating at 1/4th the speed of the C Clock (e.g., 15 MHz).

4. Memory Specifications

- **SRAM:** 16 KB of built-in Static RAM used to store data temporarily.
- **Flash ROM:** 256 KB of Flash memory used to store the application program/code.

5. Built-in Peripherals

- **Timers:** Includes two 32-bit timers (Timer 0 and Timer 1).
- **PWM (Pulse Width Modulation):** Features 6 PWM output channels, typically used for controlling motors in embedded applications.
- **GPIO (General Purpose I/O):** Contains two 32-bit ports (Port 0 and Port 1).
- **ADC (Analog to Digital Converter):** A 10-bit ADC that supports up to 4 different channels.
- **Interrupts:** Supports external interrupts and includes a Vector Interrupt Controller (VIC) to manage Interrupt Service Routines (ISR).
- **RTC (Real-Time Clock):** Used to monitor and read the current real-time and date.
- **Watchdog Timer:** A protection mechanism that resets the microcontroller if it enters an infinite loop or encounters an error condition.

6. Communication Protocols

The LPC2129 supports four main communication protocols:

- **UART (Universal Asynchronous Receiver and Transmitter):** Two available (UART0, UART1). A 2-wire protocol using **TX** (transmission) and **RX** (reception) pins.

- **SPI (Serial Peripheral Interface):** Two available (SPI0, SPI1). A 4-wire protocol utilizing **SCK** (System Clock), **MOSI** (Master Out Slave In), **MISO** (Master In Slave Out), and **SS** (Slave Select) lines.
- **I2C (Inter-Integrated Circuit):** A 2-wire protocol using **SDA** (Serial Data) and **SCL** (Serial Clock) lines.
- **CAN (Controller Area Network):** An automotive-based 2-wire protocol heavily used for device-to-device communication in vehicles, using **CAN_H** (High) and **CAN_L** (Low) wires.

▼ Microcontroller Architecture | LPC1768

1. Introduction & General Specifications

- **Microcontroller:** LPC1768.
- **Manufacturer:** Designed and developed by NXP Semiconductors (previously known as Philips).
- **Architecture Base:** It is a 32-bit microcontroller.
- **Core Processor:** Powered by the **ARM Cortex-M3**, a highly advanced and widely used processor in the industry.
- **Documentation:** To fully understand its hardware and programming, developers use the LPC1768 User Manual, which spans approximately 840 pages across 35 chapters.

2. Core System Architecture

- **AHB Matrix:** The core ARM Cortex-M3 processor connects to a multi-layer AHB matrix. This acts as a central connecting terminal or bridge linking the processor to various peripherals.
- **DMA (Direct Memory Access):** A controller that transfers data directly between devices without requiring the processor's intervention.
- **High-Speed Interfaces:** Includes built-in interfaces for Ethernet and USB device applications.

3. Clock Management & Memory

- **Clock Generation:** Utilizes an external crystal oscillator coupled with an internal **PLL (Phase Locked Loop)**. The PLL circuit generates various clock frequencies (high, moderate, or low) as needed by the system.
- **Memory Specifications:**
 - **Flash Memory:** 512 KB, used primarily for storing the application code.
 - **SRAM (Static RAM):** 64 KB, used for temporary data storage.
 - **ROM:** 8 KB.

4. Peripheral Bus Architecture (APB)

The microcontroller divides its peripherals across two separate bridges or wings to manage data flow efficiently:

- **APB Group 0 (APB0):** The first set of connected peripherals.
- **APB Group 1 (APB1):** The second set of connected peripherals.

5. Built-in Peripherals & Interfaces

- **Communication Protocols:**
 - **UART:** 4 universal asynchronous receiver/transmitters (UART0 & 1 in APB0; UART2 & 3 in APB1).
 - **I2C:** 3 Inter-Integrated Circuit interfaces (I2C0 & 1 in APB0; I2C2 in APB1).
 - **SPI & SSP:** Serial Peripheral Interface and two Synchronous Serial Ports.
 - **CAN:** Controller Area Network, widely used in automotive industries.
 - **I2S:** Inter-Integrated Sound protocol for audio applications.
- **Timers & Clocks:**
 - **General Purpose Timers:** 4 total timers (Timers 0 & 1 in APB0; Timers 2 & 3 in APB1). Used to generate delays, create interrupts, and capture timestamps of events.
 - **Watchdog Timer:** A protection circuit that resets the microcontroller if it enters an error scenario.

- **RTC (Real-Time Clock):** Manages real-time data like seconds, minutes, and dates (similar to a digital wristwatch).
- **Repetitive Interrupt Timer (RIT):** Generates continuous interrupts.
- **Analog-to-Digital & Digital-to-Analog:**
 - **ADC:** A 12-bit Analog-to-Digital Converter supporting 8 different input channels (sensors).
 - **DAC:** A Digital-to-Analog Converter used to generate physical waveforms, such as square, rectangular, or sinusoidal waves.
- **Control Peripherals:**
 - **PWM (Pulse Width Modulation):** Features 8 standard PWM channels, plus dedicated Motor Control PWM and a Quadrature Encoder.
 - **External Interrupts:** Can generate interrupts from 4 distinct external sources.
- **System Controls:**
 - **Pin Connect Block:** Allows for the configuration of different microcontroller pins.
 - **System Control:** Manages core functionalities like port pin activation and clock generation.

▼ ARM Cortex M3 Architecture

1. Introduction & Background

- **Definition:** ARM stands for **Advanced RISC Machine** (originally named Acorn RISC Machine).
- **Architecture Type:** It is based on a **RISC** (Reduced Instruction Set Computer) architecture, which utilizes simple and fewer instructions compared to complex processors.
- **Origin:** ARM is a British-based semiconductor company located in Cambridge, England, which began producing ARM processors in 1985.

2. The ARM Business Model

- Unlike typical semiconductor manufacturers, the **ARM company does not manufacture its own microcontrollers**.
- Instead, they design high-accuracy 32-bit and 64-bit architectures and sell licenses to other companies.
- Companies such as Atmel, NXP, Samsung, STMicroelectronics, and Texas Instruments buy these licenses to develop and manufacture their own microcontrollers and System-on-Chips (SoCs).

3. Core Hardware Specifications

- **Processing Power:** It is primarily a 32-bit processor, utilizing a 32-bit ALU (Arithmetic Logic Unit) and 32-bit instructions.
- **Registers:** The processor contains 16 primary registers, designated R0 through R15.
- **Clock Speed:** ARM processors typically operate at high speeds between **60 MHz to 100 MHz or more**, which is significantly faster than standard microprocessors that run at 10 to 30 MHz.
- **Buses:** It features a 32-bit address bus and a 32-bit data bus. These buses are **multiplexed** together to save space and are separated using an **ALE** (Address Latch Enable) signal.
- **Memory Capacity:** Thanks to a 32-bit memory addressing space (2³²), the ARM processor can interface with a massive **4 GB of external memory**.
- **Debugging:** It uses a **JTAG** (Joint Test Action Group) interface, allowing developers to easily debug the hardware circuit.

4. Pipelining Architecture

The ARM processor utilizes a **three-stage pipelining mechanism**, meaning it executes three distinct stages in parallel to maximize efficiency:

1. **Fetch:** The processor retrieves the specific opcode (hexadecimal instruction data) from the memory.
2. **Decode:** The fetched opcode is passed through a decoder circuit so the processor can interpret the instruction.
3. **Execute:** The ALU carries out the specified operation.

5. Operating Modes

The processor supports **7 standard operating modes** along with **1 special mode**:

- **Standard Modes:** User, FIQ (Fast Interrupt Request), IRQ (Interrupt Request), Supervisory (SVC), Undefined, Abort, and System mode.
- **Thumb Mode (Special Mode):** This is a unique **16-bit operating mode**. It is used specifically to reduce the program size and save memory space in applications with tight memory constraints.

6. Advantages & Applications

- **Advantages:** ARM processors are characterized by taking very less power, being low-cost, lightweight, and offering high accuracy.
- **Applications:** Because of their efficiency, ARM processors are extensively used in mobile phones (like iPhones), laptops, tablets, desktops, servers, and a wide variety of embedded system applications.

▼ Von Neumann vs Harvard Architecture

1. Von Neumann Architecture

- **Memory Structure:** It uses a single memory space to store both instructions and data. The single memory contains both RAM and ROM, each having its own addresses.
- **Bus System:** It utilizes only one single set of data buses and one set of address buses to communicate between the CPU and the memory.
- **Execution Style:** It operates sequentially. Because there is only a single bus system, the CPU cannot access both instructions and data simultaneously. Instead, an instruction is fetched in one cycle, and the data is fetched sequentially in another cycle.

2. Harvard Architecture

- **Memory Structure:** It uses two physically separate memories for operation: ROM is used specifically to store instructions (code), while RAM is used to store data.

- **Bus System:** It features two separate, dedicated sets of buses. There is one set of address and data buses completely dedicated to the RAM, and another distinct set of address and data buses dedicated to the ROM.
- **Execution Style:** It supports **parallel execution**. Thanks to the two different memories and separate bus systems, the CPU can access data and fetch instructions simultaneously.

3. Key Takeaway

The fundamental difference is that **Von Neumann architecture** relies on a single memory and shared buses resulting in sequential execution, whereas **Harvard architecture** separates its instruction and data memories with dedicated buses, allowing for simultaneous, parallel execution.